

AI ML TASK 4 : CLASSIFICATION EVALUATION

TASK REPORT

Questions asked :

Q1. Explain :

1. **True Positive (TP)** : The model correctly predicts a positive class.

Example:

- Patient actually has cancer.
- Model predicts cancer.

2. **False Positive (FP)** : The model predicts positive, but the actual class is negative.

Example:

- Patient does not have cancer.
- Model predicts cancer.

3. **False Negative (FN)** : The model predicts negative, but the actual class is positive.

Example:

- Patient actually has cancer.
- Model predicts no cancer.

Q2. Why accuracy alone is insufficient?

Ans. Accuracy alone can be misleading when classes are imbalanced because a model may achieve high accuracy by simply predicting the majority class. Therefore, metrics such as Precision, Recall, F1-Score, ROC Curve, and AUC provide a more reliable evaluation of classification models, especially in real-world applications like medical diagnosis and fraud detection.

Q3. Which metric is more important for medical diagnosis?

Ans In medical diagnosis, Recall (Sensitivity) is generally the most important metric because it minimizes False Negatives. Missing a patient who actually has a disease can have serious consequences, whereas a False Positive usually only leads to additional testing. Therefore, healthcare models are often optimized for high Recall while maintaining reasonable Precision.

Q4. What happens if recall is low?

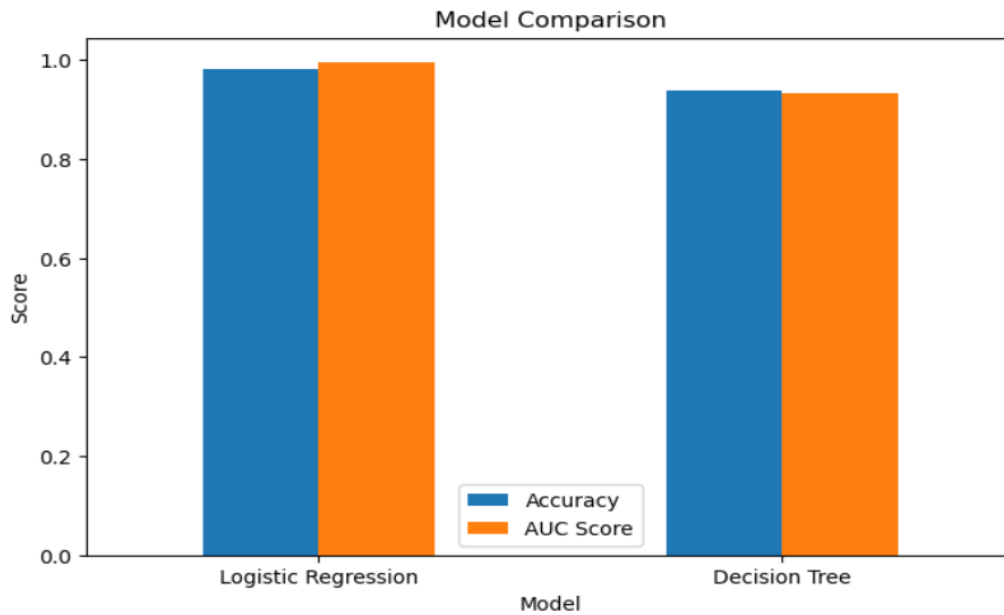
Ans If recall is low, the model produces many False Negatives, meaning it fails to identify a large number of actual positive cases. In medical diagnosis, this can be dangerous because patients with a disease may be incorrectly classified as healthy, leading to delayed treatment and serious health consequences. Therefore, high recall is generally preferred in healthcare applications.

Q. Why F1-score is preferred for imbalanced data?

Ans. F1-score is preferred for imbalanced data because it combines Precision and Recall into a single metric. Unlike Accuracy, which can be misleading when one class dominates the dataset, F1-score evaluates how well the model identifies the minority class while minimizing both False Positives and False Negatives. Therefore, it provides a more reliable measure of performance on imbalanced datasets.

COMPARISON BETWEEN LOGISTIC REGRESSION VS DECISION TREE

	Model	Accuracy	AUC Score
0	Logistic Regression	0.982456	0.995370
1	Decision Tree	0.938596	0.934193



STABILITY

Logistic Regression: More Stable

Small changes in training data usually cause only small changes in model coefficients.

Example:

- Remove a few patient records.
- Coefficients change slightly.
- Predictions remain mostly the same.

Decision Tree: Less Stable

A small change in the dataset can produce a completely different tree structure.

INTERPRETABILITY

Logistic Regression

Prediction is based on coefficients. Doctors can understand the effect of each variable.

Interpretation:

- Larger radius increases cancer probability.
- Higher texture increases cancer probability.
- Higher smoothness decreases cancer probability.

Decision Tree

Interpretation is rule-based.

Very easy to explain to non-technical users. However, very deep trees become difficult to understand.

OVERFITTING BEHAVIOUR

Logistic Regression : Because it learns a single decision boundary, it generally generalizes well. A smooth boundary separates classes. Less likely to memorize training data.

Decision Tree : A tree can keep splitting until every training sample is classified correctly. The tree may memorize noise in the training data.

Metric Selection

In this project, multiple evaluation metrics were used to assess the performance of the classification models. Although accuracy provides the overall percentage of correct predictions, it may not be sufficient when dealing with imbalanced datasets because a model can achieve high accuracy by simply predicting the majority class. Therefore, additional metrics such as Precision, Recall, F1-Score, and ROC-AUC were considered. Precision measures how many predicted positive cases are actually positive, while Recall measures how many actual positive cases are correctly identified. F1-Score provides a balance between Precision and Recall, making it useful for imbalanced classification problems. ROC-AUC evaluates the model's ability to distinguish between classes across different classification thresholds and provides a comprehensive measure of performance.

Imbalance Handling

To address the issue of class imbalance, the project utilized the `class_weight='balanced'` parameter in Logistic Regression. Class imbalance occurs when one class contains significantly more samples than another, causing the model to become biased toward the majority class. By assigning higher importance to minority-class samples, class weighting helps improve the detection of underrepresented cases and reduces the likelihood of false negatives. This approach enhances the model's ability to correctly classify minority-class instances without modifying the original dataset.

Final Model Selection

After training and evaluating both Logistic Regression and Decision Tree models, a final model selection process was carried out based on the obtained performance metrics. Logistic Regression demonstrated strong accuracy, a high ROC-AUC score, and a balanced Precision-Recall trade-off. It also showed better stability and lower susceptibility to overfitting compared to the Decision Tree model. Although Decision Trees are capable of capturing complex non-linear relationships and are easy to interpret, they tend to have higher variance and may overfit the training data. Considering its superior generalization ability, reliability, interpretability, and overall classification performance, Logistic Regression was selected as the final model for this task.

